

How to Play: Fermi Poker

Overview

Fermi Poker resembles poker, but instead of being dealt random cards, you make guesses about questions that seem unknowable at first, like “*How many dentists work in the US?*” You write down a secret guess, bet on it, get helpful hints, bet again, and win by having the best guess or by bluffing.

Before you start

- Everyone starts with the same number of chips.
- Randomly pick who acts first during the betting rounds. For the next question, the player on their left goes first, and so on.
- Don’t use phones, calculators or the internet.
- The organizer announces the ante and how often it goes up, for example every 20 minutes.

How a question is played

1. **Ante.** Everyone puts in the ante, a small mandatory bet. Then the question is revealed.
2. **Guess.** Everyone secretly writes down a range they think contains the answer, for example “400–1100”. Then betting round 1.
3. **Hint 1.** Then betting round 2.
4. **Hint 2.** Then betting round 3.
5. **Answer reveal.** Then betting round 4.
6. **Showdown.** The remaining players reveal their ranges, and the best one wins the pot. Revealing is voluntary for everyone who folded.

Your guess

- Write it secretly and place it face down. Once betting round 1 starts, you can’t change it.
- A range includes both of its endpoints.
- Write numbers however you like, for example 3,000,000, 3×10^6 , 3 million or 3M, as long as they’re clearly written and not ambiguous.

Optional game variants

- **Single numbers** instead of ranges, or **powers of ten only** (10^1 , 10^2 , 10^3 ...).
- **60-second clock:** once half the players have written their guesses, the rest have 60 seconds.
- **Mirror card** (usable once per player): write another player’s name to copy their guess.

How many dentists work in the US?

Break it down: How often do you go to a dentist? How long is an average visit? Annual working hours per dentist? US population?

Betting

Play moves clockwise. On your turn you can:

- **Check:** pass the action to the next player without betting. Only possible if nobody has bet yet in this betting round.
- **Bet or raise:** increase the highest bet. Others must match it to stay in. The minimum bet is the current ante, and a raise must be at least the size of the last bet or raise in this betting round.
- **Call:** match the highest bet so far.
- **Fold:** drop out of this question. You give up the chips you’ve put in, but risk nothing more.
- **All-in:** bet all your chips. This is possible even if what you have left is less than the highest bet.

A betting round ends once everyone has acted and matched the highest bet or folded. Within a betting round, you can only raise again if someone else raised after you.

If you’re all-in, you can win from each player only as much as you put in yourself. Extra bets go into a side pot that only the players who paid into it can win. Can’t pay the ante? You’re automatically all-in.

Who wins the pot

- The **narrowest range that contains the answer** wins. Width is the high endpoint minus the low one: if the answer is 500, a range of “200–600” (400) beats a range of “400–1100” (700).
- Equally narrow? Split the pot.
- No range contains the answer? The range whose midpoint (halfway between low and high) is closest to the answer wins.
- Everyone else folded? The last remaining player wins and doesn’t have to reveal their guess.
- Everyone, or all but one, is all-in? Since no more betting is possible, read the remaining hints and answer. Then go to the showdown.

End of the game

A player with no chips left is out. The game usually ends when one player has all the chips.